Week 2: The Hydrogen Atom
Electron Models

From the Quantum Universe to Quantum Technologies

Week 2 • Spring 2025

1. Understanding the Hydrogen Atom

1.1 Classical Behavior: Electrons in the Bohr Model

When most people think of atoms, they think of electrons rotating around a nucleus like the Earth rotates around the Sun. This is a very reasonable assumption- in fact physicists thought this was the case until 100 years ago. Let's try to understand how how orbitals work with an experiment:

`https://astro.unl.edu/classaction/animations/light/hydrogenatom.html`

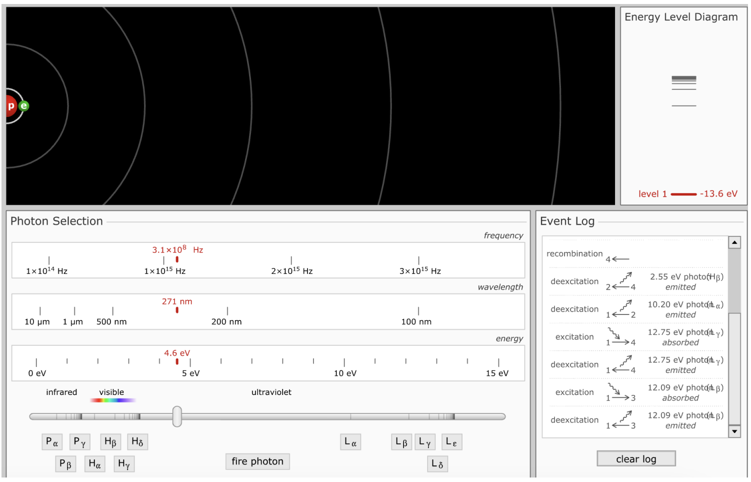

Figure 1: Model of the Hydrogen Atom

Play around for a minute by firing photons at the electron. Here we see out the electron is not randomly moving around; it is restricted to specific energy levels like rungs on a ladder.

Electrons can move to higher energy levels when they absorb energy—but only if the energy is the right amount! Imagine you're trying to ride the BART to SF and your Clipper has 4 bucks on it- even through you have money on your card you need the right amount to pay the way from Berkeley to SF.

1. Fire different energy photons at the electron: What energies excite the electron? (moves it to a higher level (temporarily))

2. How much energy does it take to ionize the electron (remove it from Hydrogen)

1.2 Quantum Behavior: Electrons as Waves

Electrons are not tiny balls orbiting the nucleus—they behave like waves. And just like musical notes or ripples in water, only certain waves fit around the nucleus in a stable way.
Imagine two people swinging a jump rope: If they move the rope at the right speed, they get a stable wave with peaks and valleys. If they shake it randomly, the wave falls apart and disappears. Electrons must exist where their waves are stable—these are the energy levels. Let's look at a sumulation of this quantum behavior.

`https://learnqm.gatech.edu/Semiconductor-Physics-Visualization/ch1/index.html`

3. Find the section where you click to try and locate the electron. Is there any pattern? Would you be able to guess where it will be before you click?

4. Find the section: "Probability vs Distance". Where do you count the most electrons? The closer to the nucleus, the farther, or somewhere in the middle?

5. Go through the next few sections: Below, draw the 1s, 2s, and 2p orbitals. Remember: the 2p orbital has three different possibilities- x,y,x- draw each.

1.3 Real Life Applications

Option A: Neon Signs. Explain in 2-3 sentences why this relates to what you learned about the photoelectric effect and energy levels. Option B: How do you think sun burns relate to photons, electrons, and the photoelectric effect? Explain in 2-3 sentences.